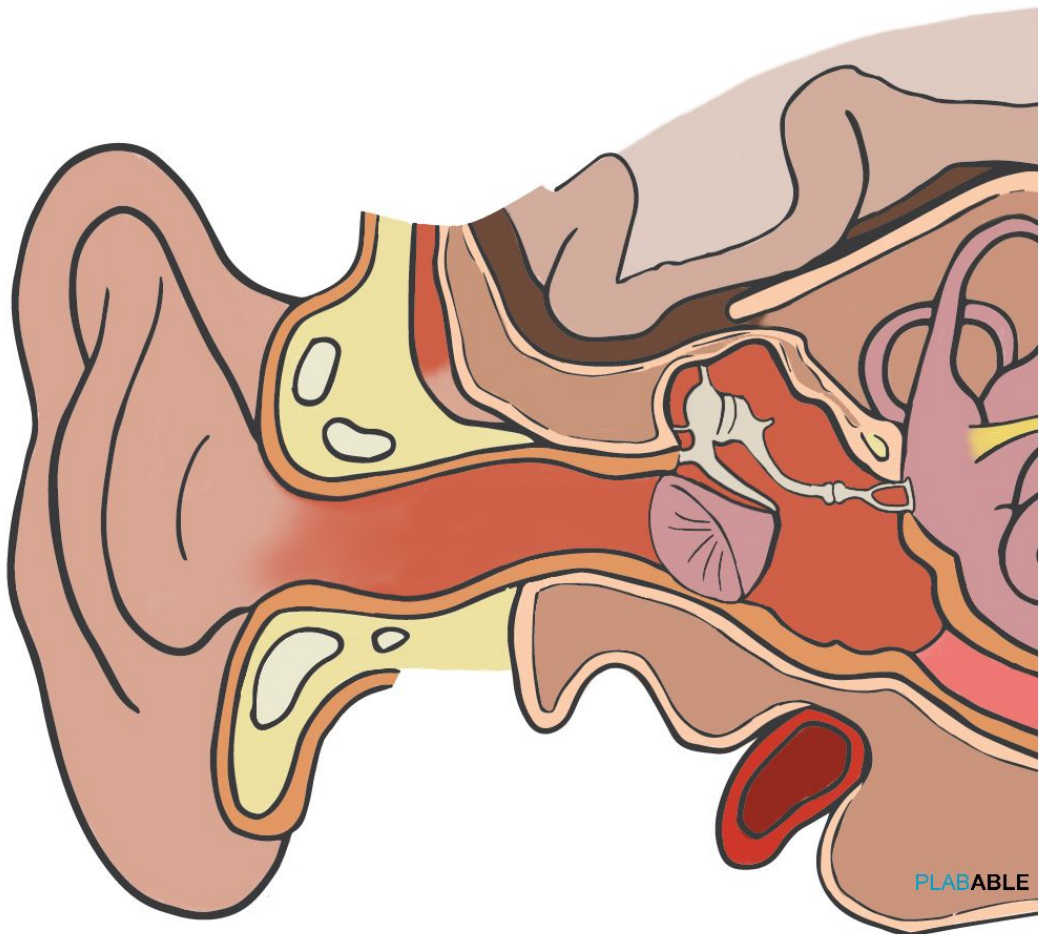


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GEMS

VERSION 1.2

ENT



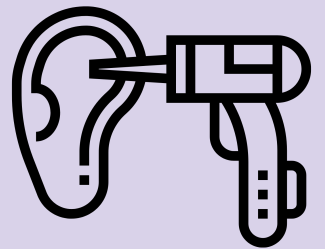
Acute Otitis Media

Presentation

- Earache
- fever
- H/o upper resp tract infection
- Pain relief followed by ear discharge - perforation

Otoscopy

- Red or cloudy TM
- Bulging of the TM
- Air-fluid level behind the TM
- Perforation and discharge (late cases)



Management

- Paracetamol or NSAIDs
- Antibiotics for children who are systemically unwell or if symptoms > 4 days
 - First line - amoxicillin for 5 days
 - Pen allergy - erythromycin or clarithromycin

Complications

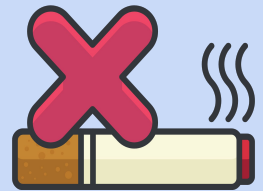
- Perforation
- Chronic Suppurative Otitis Media
- Mastoiditis
- Labyrinthitis
- Facial nerve palsy

Otitis Media with Effusion

Glue ear is a collection of fluid in the middle ear

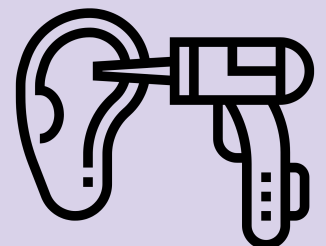
Presentation

- **Hearing loss**
- Speech or language delay
- Usually following acute otitis media
- Intermittent ear pain with fullness
- **Risk factor:** parental smoking



Otoscopy

- **Opacification of the eardrum**
- Loss of light reflex
- Retracted eardrum
- Decreased mobility of eardrum
- Presence of air bubbles or fluid level



Pure tone audiometry: Conductive hearing loss

Management

- Usually self-limiting so **reassure and review** in 3 months
- **Grommet** insertion if persistent bilaterally for 3 > months

Otitis Externa (swimmer's ear)

Presentation

- Pain and itching of the ear
- Discharge
- Conductive hearing loss (Severe cases)

Common organisms

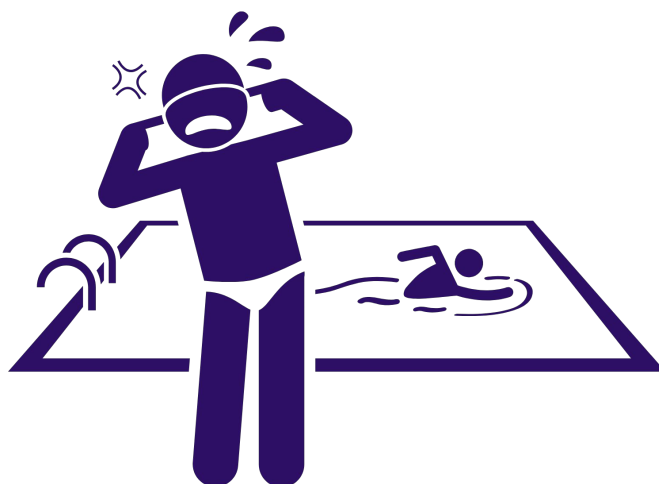
- *Staphylococcus aureus*
- *P. aeruginosa*
- Very rarely fungal

Risk factors

- Swimming
- Diabetes
- Immunocompromised patients

Management

- Acetic acid (1st week only) or
- Topical antibiotics
 - Neomycin
 - Clioquinol



Malignant Otitis Externa

Non-malignant extension of otitis externa into the mastoid and temporal bones

Presentation

- **Severe ear pain and headache**
- Purulent foul discharge
- Conductive hearing loss (Severe cases)

Common organisms

- *P. aeruginosa*
- *S. aureus*

Risk factors

- Old age
- Diabetes
- Immunocompromised patients

Management

- urgent referral to ENT (high mortality rate)

Hearing Tests (children)

Below 6 months

1. Otoacoustic Emissions (OAE)
2. Audiological brainstem responses (ABR)

6 months to 18 months

- Distraction testing

6 months to 2.5 years

- Visually reinforced audiometry (VRA)

2 - 5 years

- Speech discrimination

5+ years

- Pure tone audiogram (PTA)

Hearing Loss

Conductive

- Cerumen (Wax buildup);
- foreign body;
- debris from otitis externa;
- large exostoses,
- Osteomas
- Perforated tympanic membrane
- tympanosclerosis

Sensorineural hearing loss

- Presbycusis
- Noise trauma
- Mechanical trauma
- Meniere disease
- Medications
- Autoimmune disease
- Infection
- Neoplasm

Osteosclerosis

Features

- **Ankylosis of the stapes footplate**
- Tinnitus
- Autosomal dominant inheritance
- Seen common between 15 and 35 years of age

Investigation

- **Audiometry** - Conductive hearing loss
(Commonly bilateral)

Management

- **Medical:**
 - Hearing aids
 - Sodium fluoride
- **Surgical:**
 - Stapedectomy
 - Stapedotomy



Acute Sialadenitis

Acute infection of the salivary gland usually in dehydrated patients (e.g. elderly, post-surgery).

Presentation

- Painful swelling of gland
- Fever
- Pus leaking from duct on application of pressure

Investigation

- Oral swab (culture and sensitivity)

Management

- Antibiotics
- Good oral hygiene
- Lemon drops
- Surgical drainage

Chronic Sialadenitis (Küttner's tumour)

Chronic inflammation of the salivary gland that progresses with increasing fibrosis and parenchymal atrophy

Presentation

- Unilateral or bilateral
- Swelling, redness and pain of the salivary gland
- Most commonly **submandibular gland** is affected
- Symptoms are **more prominent when chewing food**
- **Sour taste** in the mouth
- Mouth dryness
- Localised lymphadenopathy

Investigation

- **USG**
- FNAC

Management

- Surgical resection

Cholesteatoma

Destructive and expanding keratinizing squamous epithelium in the middle ear or mastoid process

Presentation

- Chronic foul smelling ear discharge
- Progressive unilateral conductive hearing loss
- Erosion can cause facial nerve palsy and vertigo

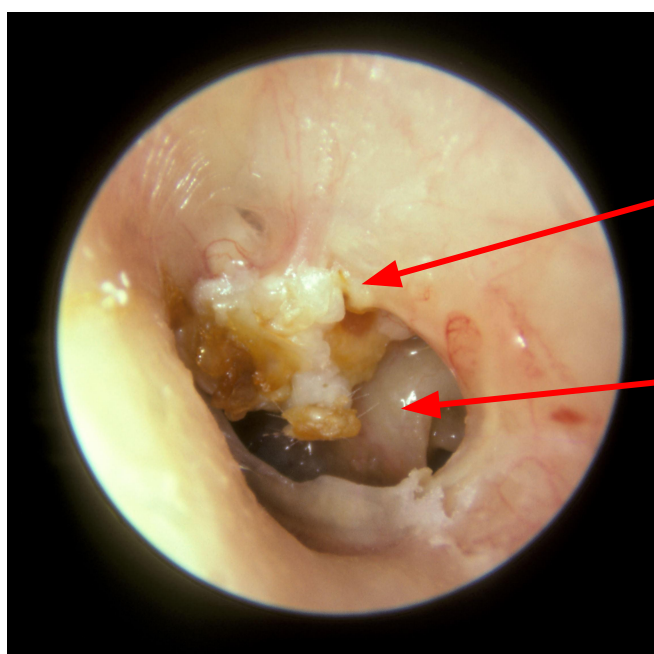
Otoscopy

- Tympanic membrane perforation
- Retracted tympanic membrane
- White mass behind tympanic membrane

CT scan - Investigation of choice

Management

- Surgery
 - **Tympanomastoidectomy (first line)**
 - Tympanoplasty



Cholesteatoma

Perforated TM

Rinne's and Weber's Test

Used to differentiate between conductive and sensorineural hearing loss

Conductive hearing loss in right ear	
Rinne's test	Weber's test
Right ear: Bone conduction better than air conduction Left ear: Air conduction better than bone conduction	Sound is localised to the right ear
Sensorineural hearing loss in right ear	
Rinne's test	Weber's test
Right ear: Air conduction better than bone conduction Left ear: Air conduction better than bone conduction	Sound is localised to the left ear

Thyroglossal Cyst

Presentation

- **Painless fluctuant swelling** in the midline of the neck
- Move upwards with swallowing and protrusion of the tongue
- Rarely cause pain and fever once it gets infected

Management

- Surgical removal



Nasopharyngeal Carcinoma

Risk factors

- Epstein–Barr virus
- Alcohol
- Smoking
- South Asian background

Presentation

- Painless swollen cervical lymph nodes
- Otitis media, epistaxis, nasal obstruction
- Conductive hearing loss, tinnitus

Management

- Urgent ENT referral (within 2 weeks)
- Biopsy (diagnosis)
- CT (staging)

Tonsil Carcinoma

Brain trainer:

A 70 year old man presents with progressive dysphagia and right sided ear pain. His right palatine tonsil is enlarged. He is a heavy smoker and drinker. What is the likely diagnosis?

→ **Tonsillar cancer**

Paranasal Sinus Carcinoma

Brain trainer:

A 46 year old man complains of a blocked nose with occasional blood present on tissues when blowing. He also complains of pressure in his upper teeth, recent cheek swelling, and double vision. What is the most likely diagnosis?

→ **Paranasal sinus carcinoma**

Throat Cancer

Brain trainer:

A 70 year old patient presents with a hoarse voice which has been ongoing for 3 weeks. He is a heavy smoker and drinker. What is on the top of your differential?

→ **Laryngeal cancer**

Nasal Polyp

Brain trainer:

A patient has asthma, rhinorrhoea, bilateral painless nasal obstruction and anosmia. What do you suspect?

→ **Nasal polyp**

Chronic Sinusitis

Brain trainer:

A 25 year old man has a headache which worsens on leaning forward. He denies nausea or vomiting. What is the most likely diagnosis?

→ **Chronic sinusitis**

Tonsillectomy Bleeding

Brain trainer:

What is the management for bleeding post-tonsillectomy?

- 1° (<24hr) → Return to the theatre
- 2° or reactionary (24hr+)
 - Admit + IV ATB

Nasal Septal Abscess

Brain trainer:

Child presents with nasal pain, tenderness, malaise and fever. He fell on his nose a few days ago but was not treated. What is the most likely diagnosis?

→ Nasal septal abscess

The fall most likely caused a septal haematoma which accounts for the temperature and malaise. The child should have had the haematoma drained a few days ago.

Perichondritis



Brain trainer:

Above patient has intense pain in his right auricle. He denies trauma or hearing loss. What is the most likely diagnosis, aetiology and treatment?

- Perichondritis
- *Pseudomonas aeruginosa*
- Fluoroquinolone

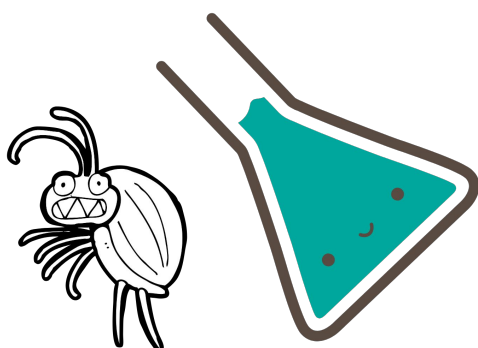
Foreign Body - Ear

- Insects to be killed with **2% lidocaine** before removal
- Adhesives (super glue) to be removed after 1 to 2 days after desquamation has occurred
- **Methods:**
 - Hook or forceps
 - **Irrigation - contraindicated for soft objects, organic material or seeds**
 - Suction - Small objects

Referral to ENT if

- Patient requires sedation
- Difficulty in removal
- Perforated tympanic membrane
- Adhesive touching TM
- After one failed attempt in a child

2% Lidocaine



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Presbycusis

Features

- **Progressive sensorineural hearing loss**
- Associated with **aging**
- Difficulty to understand in a noisy surrounding

Investigation

- Pure tone audiometry

Management

- Hearing aids
- Supportive management



Furuncle

Brain trainer:

A diabetic woman presents with a painful small, red, tender mass at the outer third of the right external ear canal. She denies hearing loss. There is no discharge. What is the most likely diagnosis?

→ **Furuncle**

Tympanosclerosis

Brain trainer:

A patient presenting with mild conductive hearing loss in left ear. Examination reveals chalky white patches on the eardrum. What is the most likely diagnosis?

→ Tympanosclerosis

Epistaxis

Features

- Either anterior (**Little's area**) or posterior
- **Trauma** is the most common cause followed by platelet disorder

Management

- Sit upright and leaning forward
- Pinch cartilaginous soft part of the nose for 10-20 min
- Still bleeds then nasal cautery followed by nasal packing

Important to remember:

- If you cannot visualise the heavy bleeding then just use nasal packing
- Vitamin K is never the correct choice in a patient taking warfarin if the INR value is not given

Nasal Trauma

Brain trainer:

A patient with a punch to the nose with a grossly deviated nose to the right. What is the most appropriate step?

→ **Speculum examination of the nasal cavity**

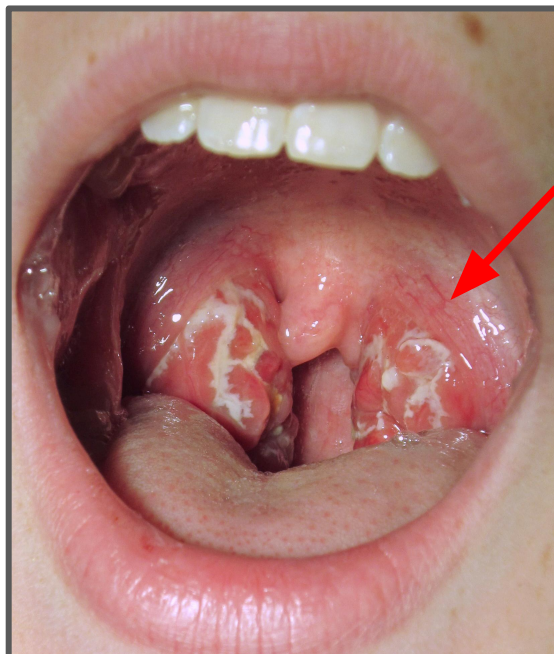
Tonsillitis

Presentation

- Throat pain
- Odynophagia
- Swollen tonsils
- Fever
- Cervical lymphadenopathy

Management

- **Symptomatic** - Paracetamol and NSAIDs
- **Antibiotics** (if severe symptoms)
 - Phenoxymethylpenicillin (1st line)
 - Clarithromycin (penicillin allergy)
- **Tonsillectomy:**
 - > 7 episodes/year for 1 year
 - > 5 episodes/year for 2 years
 - > 3 episodes/year for 3 years



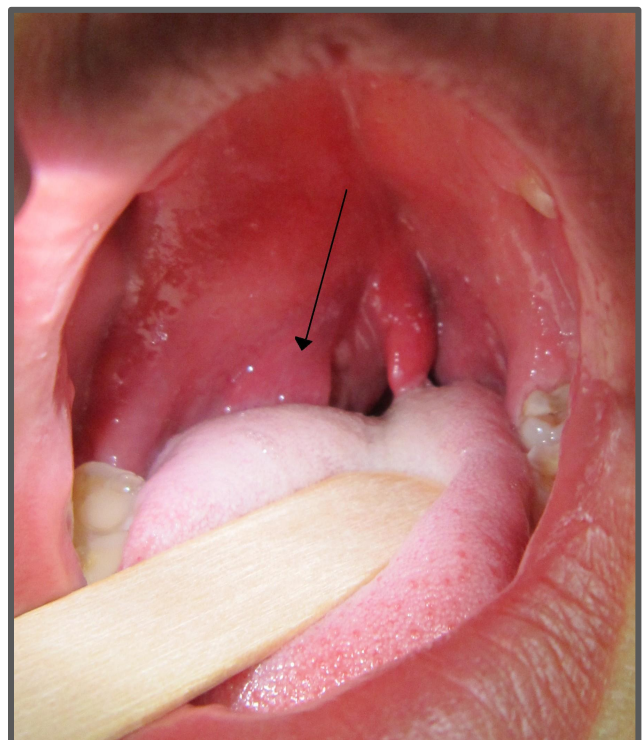
Peritonsillar Abscess - Quinsy

Features

- **Complication of acute tonsillitis**
- Swelling of the soft palate and tissues lateral to the tonsils (**peritonsillar bulge**)
- Dysphagia
- Uvular deviation
- Fever
- Trismus (Difficulty in opening the mouth)
- Altered voice - **Hot potato voice**

Management

- Aspiration
- Antibiotics (usually IV benzylpenicillin)



Benign Paroxysmal Positional Vertigo (BPPV)

Common cause of vertigo due to otoliths

Presentation

- Vertigo brought out by change in head position
- Sudden in onset
- Lasts 20-30 seconds
- Nausea
- **Dix-Hallpike** test is used to confirm

Management

- **Epley's** manoeuvre (reposition the otoliths)



Vestibular Neuritis and Labyrinthitis

Presentation

- Sudden onset and severe vertigo
- Not precipitated by head movements
- Hearing loss and tinnitus in Labyrinthitis
- H/o URI

Vertigo



H/O URI



Meniere's Disease

Presentation

- Vertigo
- Ear fullness
- Tinnitus
- Sensorineural hearing loss
- Episodic (6-11 clusters / year)

Investigations

- Audiometry
- MRI brain (To R/o Vestibular neuroma)

Management

Acute attacks:

- Prochlorperazine or cinnarizine

Prophylaxis:

- Betahistine

Acoustic Neuroma

Presentation

- **Hearing loss** (Mostly unilateral)
- Tinnitus
- Facial weakness (Facial nerve compression)
- Facial pain or numbness (Trigeminal nerve)
- Ataxia (Cerebellar compression)
- **Bilateral** acoustic neuroma occurs in Neurofibromatosis-type 2

Investigations

- MRI brain
- Audiometry

Management

Acute attacks: prochlorperazine or cinnarizine

Prophylaxis: betahistine



Pleomorphic Adenoma

Features

- Benign salivary gland tumour
- Most common gland: parotid
- Unilateral slow growing **swelling** at the **angle of mandible**
- On examination: painless, firm, mobile

Investigation

- FNAC
- Core needle biopsy (High sensitivity)
- Ultrasound

Management:

- Surgical resection



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